2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

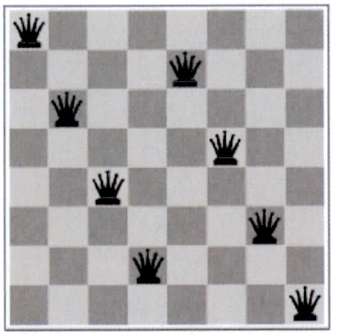
SCC.361 Artificial Intelligence

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a We discussed the 8 Queens Problem in the "search" section of the module. Below is an example of a failed state to remind you. Describe two formulations of how you can generate possible states and describe the size of the corresponding state spaces.



[4 marks]

1.b In A*-search, we have a cost function f=g+h, where g is the cost so far, and h is the estimate of the cost to the goal. What kind of search would this become if f=g (and explain why)? What kind of search would this become if f=h (and explain why)?

[4 marks]

1.c This question is about motion planning.

i. What are the inputs and outputs of a motion planning problem. [2 marks]

ii. Describe the main idea of the cell decomposition method. [1 mark]

[3 marks]

1.d You are given three variables for a Bayesian Network: "Alarm" (i.e. an alarm goes on), "Burglary" (i.e. a burglary occurs), and "Mary Calls" (i.e. a call from Mary is received). Draw your own Bayesian Network that incorporates these three variables and explain each part of your network.

[6 marks]

Question 1 continues on next page...

Question 1 continued…

1.e Take the dataset below that consists of five examples (X_1 through X_5) with three Boolean attributes (A, B, and C) and Boolean outcomes. If you build an optimal decision tree according to the Information Gain (IG) score, which attribute would be chosen as the root node of the tree? The IG score is: IG(A) = Entropy before – AvgEntropy after choosing A. You need to show your calculations up to the log() expressions (i.e. you do not have to evaluate the log() expressions), and you only need to do this for the root node of the decision tree (i.e. you can describe how you would choose which attribute as the root node without evaluating the log() expressions).

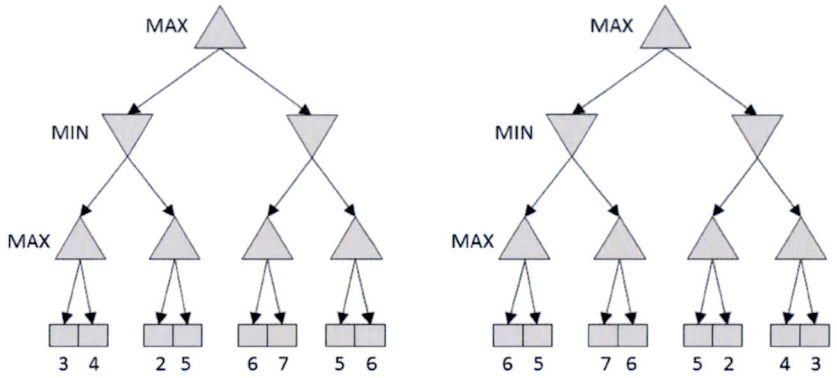
	A	B	C	Outcome
X_1	F	T	T	F
X_2	T	F	T	T
X_3	T	T	F	T
X_4	T	T	T	T
X_5	F	F	F	F

[8 marks]

Total 25 marks

Question 2

2.a Apply alpha-beta pruning to the two search trees shown below (start by copying the trees into your answer book). Indicate all parts of the trees that are not evaluated. Strike out all static values that do not need to be computed. Highlight the winning path or paths.



[6 marks]

2.b This question is about genetic algorithms.

i. Give an example of a problem where the permutation encoding would be a good choice (and explain why). [2 marks]

ii. Give an example of a problem where the value encoding would be a good choice (and explain why). [2 marks]

iii. What is elitism and what effect does it have in the overall algorithm? [2 marks]

iv. What is mutation and what effect does it have in the overall algorithm? [2 marks]

v. What is one advantage and one disadvantage of the genetic algorithm? [2 marks]

[10 marks]

2.c We discussed "CAPTCHAs" in the human computation part of the module. How can you change a CAPTCHA such that the image becomes easier for a computer to recognize?

[2 marks]

2.d This question is about iterative deepening.

i. How is iterative deepening different from depth-first search? [1 mark]

ii. Is iterative deepening complete? Explain your answer. [2 marks]

iii. Is iterative deepening optimal? Explain your answer. [2 marks]

iv. What is the time complexity of iterative deepening (and explain why)? Assume that the branching factor is b and the depth of the solution is d. [2 marks]

[7 marks]

Total 25 marks

Question 3

3.a Sketch and describe a radial-basis function (RBF) type of multilayer perceptron type artificial neural network (ANN). Stress the specific difference of RBF ANN in comparison with other types of multi-layer perceptron ANNs. **[6 marks]**

3.b Provide examples of sigmoid, linear and Gaussian type of activation functions of neurons. **[5 marks]**

3.c Sketch a single neuron of an artificial neural network and describe the analogy with the Human neuron. **[6 marks]**

3.d In the following table the expected profit of three investment decisions under different levels of traffic is given. Using Maximax, Maximin, Lagrangian (equal probabilities), and Minimum Regret Decision Making method determine which investment decision the City Council should take and what this will guarantee. **[8 marks]**

Traffic	***Expected Profit***		
	Build a bridge	*Build a tunnel*	*Use a ferry*
Low	900M	150M	£500
Moderate	250M	£300M	£500
High	80M	£500M	£400

Total 25 marks

Question 4

4.a Describe and provide an example of each of the following learning paradigms:

i. Supervised learning [3 marks]
ii. Unsupervised learning [3 marks]
iii. Supervised learning with reinforcement [3 marks]

[9 marks]

4.b Describe and sketch the membership function of the fuzzy set 'cold' and 'warm' centred at 0 and 20 degrees C respectively using:

i. triangular membership functions **[2 marks]**

ii. Gaussian membership function **[2 marks]**

iii. trapezoidal membership function **[2 marks]**

4.c For the following two data samples:

$x_1=\{12; 35; 31; 62\}$
$x_2=\{22; -38; 62; 50\}$

i. calculate the range and mean **[4 marks]**
ii. perform normalisation **[2 marks]**

4.d Compare the sets of 'odd numbers' and 'young age'. Which of the two is fuzzy and which is a classical/traditional set. Why? What is the difference between the two?

[4 marks]

Total 25 marks

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